

Roll No.

--	--	--	--	--	--	--	--

Paper Code: TCS-401

10/1/19
1.30-4.30

END Semester Back paper Examination December- 2018

Course- B.Tech (CS) - IV Sem

Paper Name – Computer Based Numerical and Statistical Techniques

Time: Three Hours

MM: 100

Note:

- I. This question paper contains five questions.
- II. All Questions are compulsory.
- III. Instructions on how to attempt a question are mentioned against it.
- IV. Total marks assigned to each question are twenty.

Ques 1.

(Attempt any two questions of choice from a, b and c)

(10 X 2= 20 Marks)

- a. Write an algorithm to find roots of a quadratic equation, and find the product of two with all the possible conditions. Example: $F(x)=x^2+5x+6$
- b. Describe how the rate of convergence of two methods be compared, explain by comparing Bisection and Regula-falsi method for finding the roots of an equation.
- c. Describe the following:
 - i) Types of errors.
 - ii) Underflow and overflow conditions of error in floating point addition and subtraction.
 - iii) Rounding off error
 - iv) Propagation of error

Ques 2.

(Attempt any two questions of choice from a, b and c)

(10 X 2= 20 Marks)

- a. The following table gives the marks obtained by students in Statistics

Marks	Number of students
10-20	128
20-40	144
40-50	110
50-60	116
60-70	172
70-80	189

Use Newton's forward formula to find the number of students who got more than or equal to marks.

- Find the root of the equation $F(x): x^3 - x - 1 = 0$, lies in the interval (1, 2). Perform three iterations of the bisection method.
- Find the root of the equation $F(x): 2x^3 - 4x - 10 = 0$, lies in the interval (2,3) using Regula-Falsi method with at least four iteration.

Ques 3.

(Attempt any two questions of choice from a, b and c)

(10 X 2 = 20 Marks)

- The function $y=f(x)$ is given at the point (7,3), (8,1), (9,1) and (10,9). Find $f(9.5)$ using Lagrange's formula.
- The following data is taken from the stream table

Year (Yr)	Population (p) in million
1914	17.899
1915	18.154
1916	20.756
1917	22.288
1918	24.322

By using newton backward difference interpolation formula find the population in the year $y=192.1$

- Explain Finite Difference in Interpolation, Calculate finite difference table of the data given below

Product sell month	Sale (S) (in \$)
Jan	14.51
Fab	16.85
Mar	18.644
April	20.98
May	22.211

Ques 4.

(Attempt any two questions of choice from a, b and c)

(10 X 2 = 20 Marks)

- Compare Trapezoidal and Simpson 1/3 methods. Evaluate the following integral equation by Trapezoidal rule.

- b. What is the importance of forward difference and backward differences in polynomial interpolation, explain with suitable example.
- c. Evaluate the integral by Simpson 3/8 rule, by considering suitable intervals..

Ques 5.

(Attempt any two questions of choice from a, b and c)

(10 X 2 = 20 Marks)

- a. Write an algorithm for polynomial FIT. Find a second degree polynomial to the following data.

X	1.0	2.0	3.0	4.0
Y	3.0	5.0	9.0	15.0

- b. Given the following data:

X	1	5	3	2	1	1	7	3
Y	6	1	0	0	1	2	1	5

Fit a regression line of y on x.

- c. Describe the following:
- Continuous frequency distribution
 - Histogram
 - Frequency polygon
 - Frequency curve
-